

Exercise on Stored Procedures

Consider the following database for a sales system, where information about customers, products, orders, and payments is stored and maintained. The Customers table stores details of customers who place orders (*Customers: CustomerID, Name, City*). The Products table maintains information about items available for sale (*Products: ProductID, ProductName, Price*). The Orders table represents purchase transactions made by customers (*Orders: OrderID, CustomerID, OrderDate*). The OrderDetails table stores detailed information about each order (*OrderDetails: OrderDetailID, OrderID, ProductID, Quantity*). The Payments table Tracks payment information for orders (*Payments: PaymentID, OrderID, AmountPaid*). The database includes the following relationships: a customer can place multiple orders, each order belongs to one customer, an order can contain multiple products (via OrderDetails), A product can appear in many orders, and each order can have one or more payments.

- a) Create the database by including all the described relations and relationships with appropriate primary and foreign key constraints.
- b) Create stored procedures for each of the following:
 1. To get the customer details for a given customerID.
 2. To get the orders by customerID.
 3. To get the product details by productID.
 4. To get the payments by orderID.
 5. To get the customer details by city.
 6. To get the product details which are having the price above the given value.
 7. To get the orders for a given date.
 8. To get the order details for a given product.
 9. To get the payment details above the given amount.
 10. To get the order details within the given range.
 11. To get the customer count as the output.
 12. To get the total payment as the output.
 13. To get the maximum product price.
 14. To count orders per customer.
 15. To get the average price.
 16. To get the minimum payment.
 17. To get the total quantity.

18. To count the order details.
19. To update the payment for an order by a given amount.
20. To apply a discount for a given product.
21. To increase the price of a given product by a given amount.
22. To adjust the quantity for a given order.
23. To add 10% tax for a particular payment above the given threshold.
24. To update the city of a customer with the given value.
25. To modify the price of a product by the given value.
26. To calculate the total revenue per customer and output it.
27. To get the customer order count by city.
28. List down all the stored procedures created for the sales database.